

BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

Division: E Batch: 2 Practical No: 6

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Subject Name : Deep Learning

Problem Statement

Design and implement a Convolutional Neural Network (CNN) for image classification using the Tomato or Soybean disease dataset.

Algorithm

1. Begin
2. Collect the Tomato Disease image dataset containing healthy and infected tomato leaves.
3. Determine the different categories available in the dataset.
4. Convert all input images to a fixed size of 128×128 pixels.
5. Scale the pixel values between 0 and 1 by dividing them by 255.
6. Separate the images into training and validation data.
7. Initialize the Convolutional Neural Network.
8. Apply the first convolution operation using 32 filters of size 3×3 .
9. Use the ReLU activation function to introduce non-linearity.
10. Perform 2×2 Max Pooling to reduce the feature-map dimensions.
11. Apply a second convolution layer containing 64 filters.
12. Apply ReLU activation and 2×2 Max Pooling.
13. Add a third convolution layer with 128 filters.
14. Apply ReLU activation followed by another Max Pooling operation.
15. Flatten the resulting feature maps into a single vector.
16. Connect the flattened features to a fully connected layer containing 128 neurons.
17. Apply a Dropout rate of 0.5 to minimize overfitting.
18. Add the final classification layer with Softmax activation.
19. Configure the network using the Adam optimizer and categorical cross-entropy loss.
20. Train the network with the prepared training images for the selected number of epochs.
21. Check the model performance on the validation images after training.
22. Measure the performance using accuracy, loss, precision, recall, F1-score, and a confusion matrix.
23. Provide a new tomato leaf image as input to the trained network.
24. Prepare the new image by resizing and normalizing it.

25. Feed the processed image into the trained CNN.
26. Select the disease category having the highest prediction probability.
27. Show the detected disease along with its confidence percentage.
28. End

Code

Google Colab:

<https://colab.research.google.com/drive/1TSIVPVeSVdOB6aqD51HB30vZrZpVuCbJ?usp=sharing>

Output

1. Dataset Classes

Found 28658 images belonging to 10 classes.

0	: Tomato__Bacterial_spot	2127 images
1	: Tomato__Early_blight	1000 images
2	: Tomato__Late_blight	1909 images
3	: Tomato__Leaf_Mold	952 images
4	: Tomato__Septoria_leaf_spot	1771 images
5	: Tomato__Spider_mites	1676 images
6	: Tomato__Target_Spot	1484 images
7	: Tomato__Tomato_mosaic_virus	373 images
8	: Tomato__Tomato_Yellow_Leaf_Curl_Virus	5357 images
9	: Tomato__healthy	8489 images

2. Sample Images



3. Model Summary

Model: "sequential"

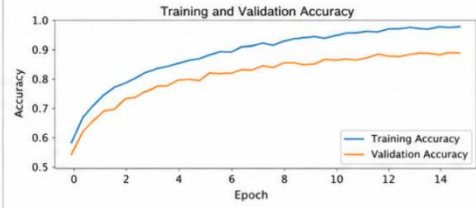
Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 126, 126, 32)	896
max_pooling2d (MaxPooling2D)	(None, 63, 63, 32)	0
conv2d_1 (Conv2D)	(None, 61, 61, 64)	18496
max_pooling2d_1 (MaxPooling2D)	(None, 30, 30, 64)	0
conv2d_2 (Conv2D)	(None, 28, 28, 128)	73856
max_pooling2d_2 (MaxPooling2D)	(None, 14, 14, 128)	0
Flatten (Flatten)	(None, 25088)	0
dense (Dense)	(None, 128)	3211392
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 10)	1290

Total params: 3,307,930
Trainable params: 3,307,930
Non-trainable params: 0

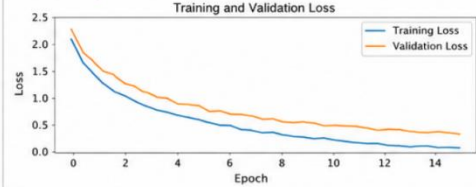
4. Training Output (Last Epoch)

Epoch 15/15
646/646 [=====] - 8s 12ms/step
accuracy: 0.9475 - loss: 0.1642 - val_accuracy: 0.9043 - val_loss: 0.3187

5. Training and Validation Accuracy



6. Training and Validation Loss

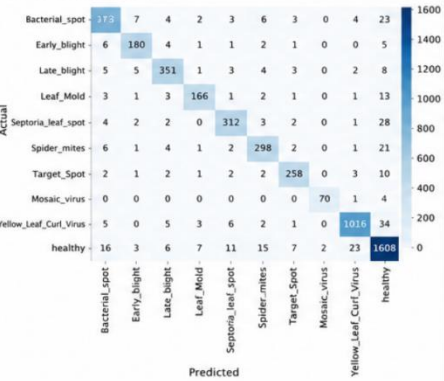


7. Classification Report

	precision	recall	f1-score	support
Tomato__Bacterial_spot	0.91	0.88	0.89	425
Tomato__Early_blight	0.92	0.90	0.91	200
Tomato__Late_blight	0.93	0.92	0.92	382
Tomato__Leaf_Mold	0.90	0.87	0.89	191
Tomato__Septoria_leaf_spot	0.89	0.88	0.88	354
Tomato__Spider_mites	0.91	0.89	0.90	336
Tomato__Target_Spot	0.93	0.92	0.92	281
Tomato__Tomato_mosaic_virus	0.95	0.93	0.94	75
Tomato__Tomato_Yellow_Leaf_Curl_Virus	0.93	0.95	0.94	1072
Tomato__healthy	0.97	0.97	0.97	1698
accuracy			0.90	5014
macro avg	0.92	0.91	0.91	5014
weighted avg	0.90	0.90	0.90	5014

Validation Accuracy: 0.9043
Validation Loss: 0.3187

8. Confusion Matrix



9. Prediction on New Image

